Linear Regression

Equation $Y = mx + b$

Y = Dependent Variable

X = Independent Variable

m = Slope (how much Y changes ~~fro~~ for for a unit change in X)

b = Intercept (Value of Y when X is 0)

Project : House Price Predictions

Sqft	Price (lacs)
1200	120~~000~~
800	80
500	55

Sqft (X)	Price (Y)	Mean (X)	Mean (Y)	Deviations (X)	Dev (Y)
1200	120	833.33	255	-366.67	135
800	80			33.33	175
500	55			333.33	200

product of Dev (X × Y)	Sum of product dev	Square of ~~Dev~~ for X
-49500.45	22998.3	134446.88
5832.75		1110.88
66666		111108.88

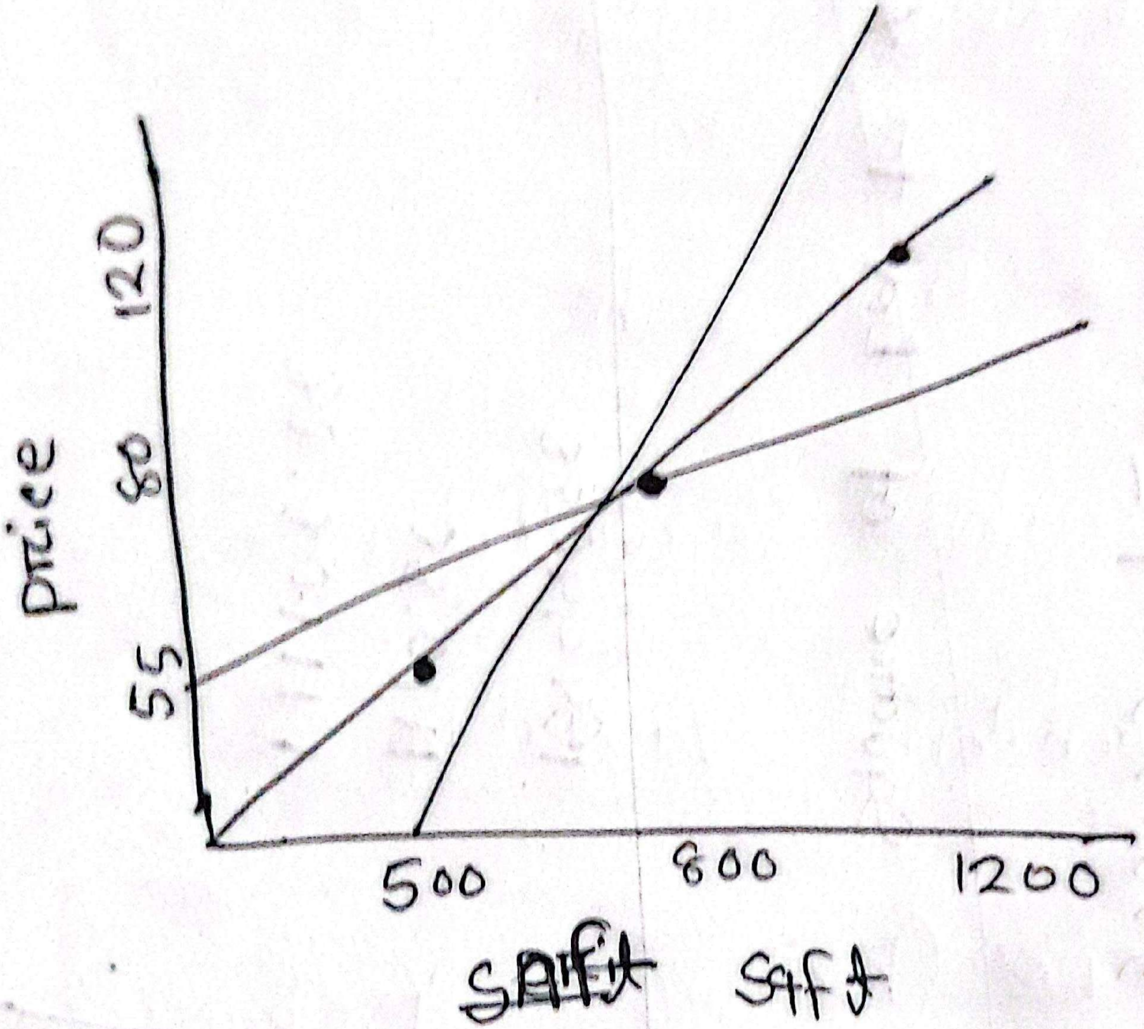
m = Sum of product of dev / Sum of square of dev for X

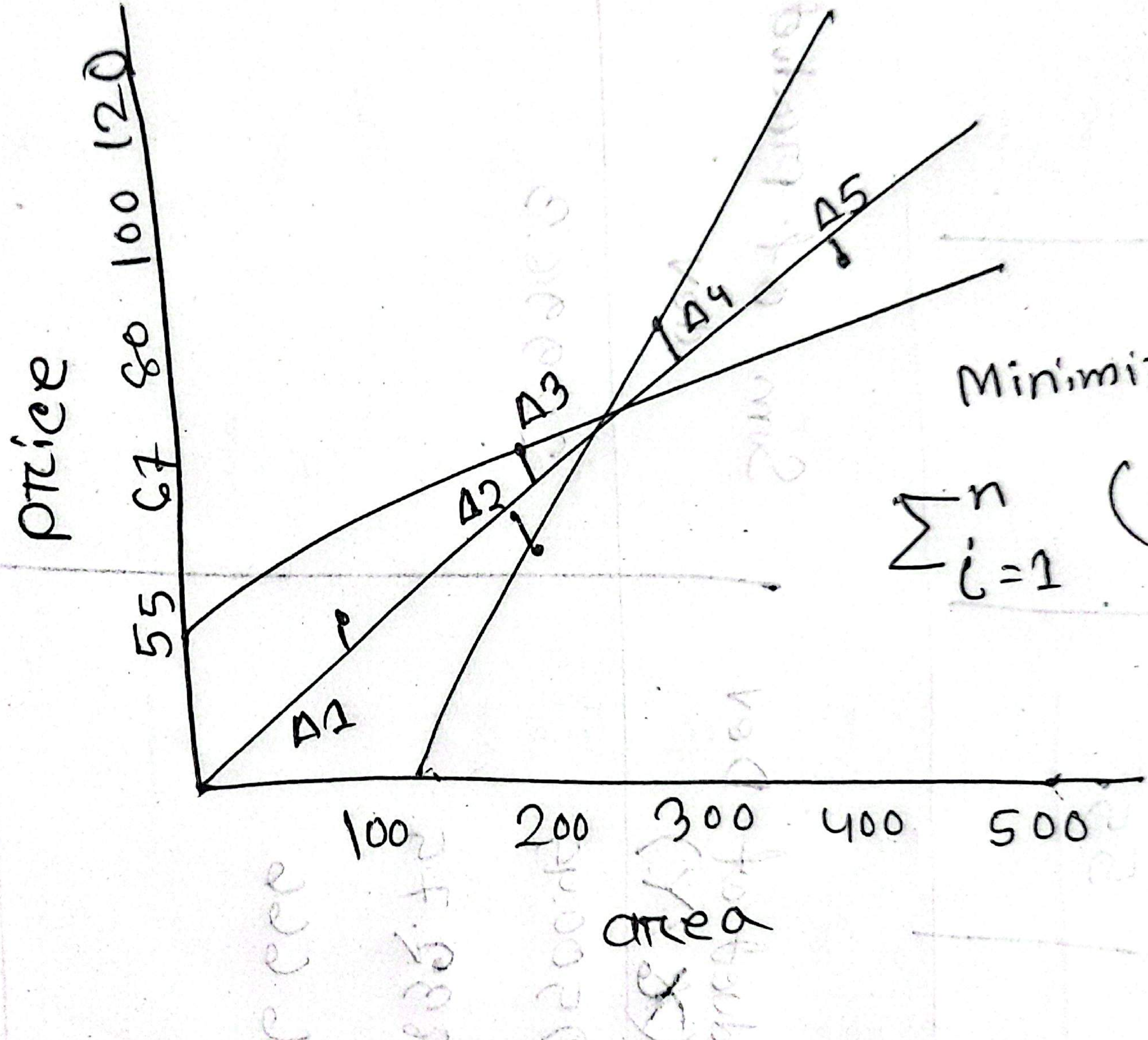
b = Mean of Y - (m * Mean of X)

We know that,

$$y = mx + b$$

$$\Rightarrow b = y - mx_0$$


price
120
80
55
500
800
1200
Sqft



Minimize

$$\sum_{i=1}^{n} (\Delta i)^2$$